

Ethical Considerations in the use of big data in Healthcare: Privacy and Ownership as a case study

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Abstract. The healthcare industry is made up of a massive amount of data that is typically built from different sources, including electronic health records, medical research, wearable devices, patient's portals, search engines among others. Therefore, the aim of this research is to highlight the issue and strategies to control the problem of patient privacy and data ownership in the age of big data in healthcare. Patient privacy in the context of big data has been an issue faced in healthcare, but data ownership is a relatively new issue which is yet to be fully recognized in the healthcare sector. This study will be focusing on ethical issues and the governance and compliance strategies to the problem of privacy and data ownership in the healthcare system.

Keywords: Big data, ethics, privacy, ownership, healthcare.

Contents

1	INTRODUCTION	3
2	BIG DATA IN HEALTHCARE.....	4
2.1	Artificially Built Data	5
2.2	Biometric Information	5
2.3	Data Produced by Humans.....	5
2.4	Data from Transactions.....	5
2.5	Medical Information	5
2.6	Information from Publications.....	5
2.7	Volume	5
2.8	Velocity.....	6
2.9	Variability	6
2.10	Variety	6
2.11	Veracity.....	6

3	ETHICAL FRAMEWORKS FOR THE USE OF BIG DATA IN HEALTHCARE.....	6
4	PATIENT PRIVACY AND DATA OWNERSHIP AS ETHICAL ISSUES OF BIG DATA IN HEALTHCARE.....	7
5	GOVERNANCE AND COMPLIANCE STRATEGY.....	9
6	CONCLUSION	10
7	BIBLIOGRAPHY	11

1 INTRODUCTION

Systematic transformation is currently taking place in the world's health care systems. We have reached a critical turning point when we can significantly change how information is generated and accessed. We are now able to organize and modify the techniques by which we handle this information as a consequence of the rise and growth of big data as a result of the digital revolution, and as a result, in the field of health care, the power to control and cure illnesses. On the other side, the advent of big data in healthcare offers new difficulties, particularly in terms of safeguarding, ownership, privacy and confidentiality.

Big data analytics can raise the risk of privacy breaches through the acquisition, protection, and usage of patient data. El Emam and Jonker discovered in a study that researchers can often identify individuals in datasets that are thought to be anonymous by cross-referencing with other publicly available information" (El Emam and Jonker, 2016). Furthermore, without the patients' express permission, the use of machine learning algorithms can reveal private information about them (Mittelstadt and Floridi, 2016).

The privacy and confidentiality of patients should be protected with serious caution , because concerns about privacy and confidentiality limit the ability to link external data to a person's insured data in ways that could otherwise personalize services and improve consumer experiences with healthcare(Yuen-Reed & Mojsilovi, 2016). Because so much health care data is centralized, it is especially vulnerable to misuse. (Mohr et al., 2013).

Healthcare providers must make sure they have strong data security mechanisms in place to protect patient data in order to meet these issues. Utilizing encryption and secure data storage techniques, granting only authorized personnel access to patient information, and ensuring that all staff members are trained in data security protocols are some examples of how to do this.

Big data has opened several opportunities for the healthcare industry. In particular, the use of big data in healthcare has reportedly resulted in the standardization of information technology (IT), the application of evidence-based and value-conscious medicine, enhanced preventative care, and personalized therapy. (Harvard Business Review, 2014). Big data helps with patient diagnosis as well as treatment by seeing indications and trends in the data. Big data has enhanced the treatment of patients while cutting costs and increasing patient happiness in the healthcare industry. However, there are a number of challenges with using big data in healthcare, including safeguarding and privacy concerns. This is because erroneous input, data abuse, identity theft, and unauthorized distribution are all quite common with electronic health records. (Iroju and Ikono, 2013).

The concept of data ownership is significant because it means some level of control over how personal health data is utilized (Beitz, 2014). Other researchers who have looked at how property law and privacy law interact with regard to medical records and

health data have come to the conclusion that none of these laws fully apply to health information or a patient's capacity to control their health information. (Radin, 1982).

Given the importance of the subject, it is noteworthy that there isn't much academic dialogue in this area (Koskinen et al., 2012). Most of the research on this topic can be found in the area of jurisprudence, but certain researchers known for their writings in ethics and health care also bring up this problem. (Lessig, 2012).

When dealing with patient-generated data, like as information obtained by wearable technology or patient portals, the issue of data ownership is particularly crucial. Patients could think they own this information and have control over how it is utilized. However, healthcare professionals could assert ownership of this information by saying that it is necessary for research.

Patient data should be exclusively owned by patients themselves in order to boost confidentiality and privacy, but some data must be publicized in order to protect the public interest, including data related to important governmental organizations that contribute to public health, individual and patient safety and medical research. (Rodwin, 2010).

This study examines the concept of big data in the context of healthcare, using patient privacy and data ownership as focus points. The key ethical challenges of big data in health care will be discussed as well as governance and compliance strategies to these ethical challenges.

2 BIG DATA IN HEALTHCARE

Big data is defined as a sizeable amount of fast-moving, complicated, and changing data that calls for contemporary methods and technology for information collection, archiving, distribution, management, and analysis (Yaseen & Obaid, 2013). In order to economically extract value from huge amounts of various types of data, big data as an emerging form of technology enables high-velocity collection, exploration, and evaluation (Burghard, 2012). Big data is a term used to describe a set of massive and intricate electronic health records that are challenging to manage, disseminate, and evaluate using traditional techniques. Big data is also a term used in the healthcare industry to describe a collection of instruments and technology that are used to efficiently produce, store, process, examine, and retrieve enormous volumes of electronic health information. Big data is used by data professionals in the healthcare industry for a variety of purposes, including enhancing patient care and creating sophisticated models using machine learning that can identify medical issues from x-ray images. Data professionals utilize analytics as a tool to manage and evaluate enormous volumes of data in order to offer insights, spot patterns, and support decision-making.

Electronic health records, which usually include a patient's medical history, demographic data, prescriptions, immunization records, test results, and progress reports, are the most important source of big data in the healthcare industry. Patients can receive the best care possible because to the adoption of electronic health records, which have replaced the time-consuming, hard-to-share, and sometimes unreadable

handwritten files that were previously used to record this information. However, the term big data refers to both large data volumes and the capacity to apply cutting-edge approaches and technologies to draw out meaningful information from a huge collection of data sets. Some sources of big data in healthcare includes:

2.1 Artificially Built Data

These figures were compiled using equipment used in the healthcare industry. Examples include data from smart meters, vital sign monitors and data from wearable devices.

2.2 Biometric Information

Patient's physical attributes, such as data from fingerprinting, genetics, signatures, retinal scans, x-rays, and other medical images, are used to generate these data.

2.3 Data Produced by Humans

These comprise information produced by people within the healthcare system. medical research, wearable devices, patient's portals fall under this category. Another type of data produced by humans is data from electronic health records.

2.4 Data from Transactions

These are made up of data from medical service billing and claim records.

2.5 Medical Information

Vital statistics, health surveys, and disease registers are some of these data sources.

2.6 Information from Publications

These include data from clinical studies and information from journals.

Big data exists to significantly improve healthcare. The question what is big data is raised, and why is it important? The three Vs that define big data—large amounts of data (volume), quick access to and analysis of that data (velocity), and significant data disparity among individuals and data types (variety)—are all present in medical data (Sun and Reddy, 2013). This study classifies big data in healthcare into 5V's due to the alteration in the context of healthcare data:

2.7 Volume

Volume is used to describe the size and amount of data produced by healthcare industry players.

2.8 Velocity

In order to treat patients effectively and efficiently, the healthcare sector requires the interchange of health data to be smooth, safe, and beneficial. This is because patients usually receive treatment from a wide range of healthcare professionals in different parts of the world. To make decisions, healthcare practitioners need immediate access to this data (Iroju et al., 2013). Consequently, velocity is necessary in health industry since it enables healthcare practitioners to share and utilize data quickly.

2.9 Variability

Data variability can be seen in the velocity, format, framework, and interpretation of the data. Data from the healthcare industry is available in many different formats, including simple files, relational database systems, visuals, and texts. A term's definition may change within the healthcare industry. For instance, in radiography, a breast mass is referred to as breast cancer whereas a lung tumour is referred to as a mass. (2006) Friedman and Johnson.

2.10 Variety

The healthcare system is made up of several resources of data, including data that is structured, semi-structured, and unstructured resources. Numerous resources, including digital and social media, internet host logs, and monetary transactions are used to collect data about healthcare. Healthcare providers may also gather data from many sources and keep it in a range of traditional and contemporary technologies and numerous databases.

2.11 Veracity

Measurement of data generation speed is called veracity. A major cause of worry has been the type of data generated by the health care sector (Feldman et al., 2015). The standard of data created within the healthcare system is vital because judgements that could determine the differences between life and death depend on accuracy and excellence of healthcare data which help medical professionals in making crucial decisions at the correct moment. Therefore, health information must be transparent, correct, and applicable.

3 ETHICAL FRAMEWORKS FOR THE USE OF BIG DATA IN HEALTHCARE

Choosing how to utilize big data involves a variety of complex ethical dilemmas. Some of these issues are prevalent in more conventional data ecosystems as well, but in the context of big data they are either more urgent or the old methods for tackling them are no longer useful. The Singaporean initiative for science, health, and policy-relevant

ethics (SHAPES) designed the current framework. The complicated ethical concerns relating to the use, sharing, and reuse of big data in health and scientific studies are covered in this framework. It specifically includes patient's privacy and data ownership as ethical concerns of big data in health care.

It is essential to create an ethical framework for big data in healthcare in order to preserve patient privacy and data while maximizing its potential benefits. An ethical framework, which offers principles for responsible data use and decision-making, supports legislative frameworks like HIPAA and GDPR by providing a foundation for privacy protection. Key components of an ethical framework for big data in healthcare include the following: privacy, informed consent, data anonymization and de-identification, safeguard and data governance, transparency and accountability, beneficence and non-maleficence.

Data for Humanity: An Open Letter is the title of an essay written by two academics, Robert V. Zicari and Andrej Zwitter, that was posted on the website of the Goethe University in Frankfurt (Big Data Lab). This framework discusses the experience of big data's remarkable rise, the advancement of data science, and the growing pervasiveness of data analytics. The approach emphasizes both the benefits and drawbacks of organizations of all sizes routinely acquiring personal data. It aims to bring together experts from various fields to promote the responsible use of data so that it can benefit humanity. Its guiding principles are that data usage should not harm anyone; that it should be used to foster peaceful coexistence; that it should be used to assist those who are vulnerable or in need; that it should be used to protect and enhance the natural environment; and that it should be used to help create a just and inclusive society.

4 PATIENT PRIVACY AND DATA OWNERSHIP AS ETHICAL ISSUES OF BIG DATA IN HEALTHCARE

There are a quite number of key ethical issues in the use of big data in healthcare like Informed consent, bias, privacy, ownership, data security. But this study will concentrate on patient privacy and data ownership as a case study.

Health care data is more private and controlled than other kinds of Big Data. A key source of concern is privacy and confidentiality (Mohr et al., 2013). But there is currently no ideal response to the problem of safeguarding patient data privacy. Patient data leaks may result in unintended consequences (such as damage, prejudice, and other things). Both nationally and globally, there are many actual examples. The use of big data technologies increases the risk to personal medical data. In the Big Data age, some even believe that it is impossible to safeguard individual privacy (Schadt, 2012). However, since the identification and elimination of identification of data still need people or applications to handle personal data, the likelihood that a patient's health data may be misused by others without their permission or consent increases (Rothstein, 2010). Eliminating identities and digital identity encryption may help to curb the problem. Big data increases the dangers to patient data for two reasons. First, the data's own danger. The capacity to copy and maintain data without respect to time or location is characterized by high risk and long-term risk under big data settings. The other threat

is posed by Big Data technologies. Although a big data source adopts unidentifiable, personally identifiable and encrypted data, residual risk can still identify a user identity and data connection technologies can re-determine individual identities (Ward, 2014). Big Data adopts pseudonymized, private, personal data that has been anonymized but still has a small risk of identification that is when numerous data are used to relate, the risk increases. De-anonymization is a strategy used in attempts to identify the anonymous data sources by comparing the anonymous data with data from other sources (Yom-Tov, 2016).

Ownership is a nuanced idea that encompasses the freedom to modify, redistribute, and profit from intellectual property and innovations resulting from the data analysis. In order to maintain data integrity, the data owner may impose limitations on the redistribution and modification of the data, but data operators and analysts may still have access to the data for analysis, invention, and the creation of intellectual property. Even though they are the legal owners of their own health data, which may be kept and managed by hospitals, doctors, medical laboratories, GPs, pharmacies, and governmental agencies in countless, incompatible data storage compartments, consumers who need medical care may not have access to or control over that data. The ownership notion, according to some participants on both sides of this debate, is essential for resolving the ethical problems around access to health data. There are others who support the privatization of data (Privatization Postulate) who believe that private ownership involves a person's freedom to manage their data, privacy, and property (Hall and Schulman, 2009). In other view, proponents of open science argue that everyone should have access to personal health information (Rodwin, 2009). The current craze is The Communization Postulate, commonly known as Public Ownership.

The privatization postulate and the communization postulate both allude to core moral principles like fairness and autonomy as well as the apparent ease of ownership, which may make them both seem desirable. They appear to be an answer to a complicated problem, but they also bring up moral, intellectual, and practical concerns. The argument between these two diametrically opposed points of view diverts attention from other ideas that take a more nicety viewpoint to ownership and have recently been addressed, such as data cooperatives, public data trusts, effective information sharing conducts, and the concept of private data power, as well as from creating policies that protect individual's privacy (Blasimme et al., 2018). There are various concepts of ownership, though. From the law point of view, partitioning of ownership is possible, and various parties may be permitted to utilize certain aspects of an item. Party B may have the right to reside there till she dies away even if Party A may be the owner of a certain piece of property. It implies that B's rights constrain A's choices, and that B will retain the right to inhabit the property even after A sells it to C. Even more complicated forms of ownership include things like corporate shares, royalties, etc.

Private ownership model exists mainly to provide individuals, the ability to govern data. It indicates that an individual is free to disclose or sell their personal information to their own healthcare provider or to third parties, such as databanks or other institutions. Embedded research cannot be properly conducted without a comprehensive dataset, and conclusions about a provider's performance may be made that are not valid. Furthermore, these inaccurate conclusions run the danger of

damaging the provider's reputation and prompting patients to lose trust in them if they are unfavorable. As a result, a more flexible approach is needed. This problem is avoidable, and institutional healthcare providers have several practical options for minimizing reputational concerns.

Both an economic argument, which sees knowledge as a specific form of good that should be available to everyone, and a moral argument, which emphasizes the advantages for the healthcare sector, are used to support the communization postulate (Hess and Ostrum, 2011). The communization postulate for individual-level health data calls for the development of a system that would make it possible to sustainably manage the communal use of the infrastructure. These practices create enforceable standards to impose a certain level of uniformity, safeguard and promote the creation of common products. This viewpoint's proponents believe that a managed commons should provide information to all stakeholders in order for them to make informed decisions and have a method for resolving conflicts that ensures adherence to social norms. This system as a whole has to be flexible enough to adapt to changes in the way information is stored, processed, and sent. This system includes any necessary infrastructure (Hess and Ostrum, 2011). With all of these elements in place, scientific knowledge should be a safe shared resource. Open-access publications, public library systems, global and national research organizations, and research programs are now constructing this system.

5 GOVERNANCE AND COMPLIANCE STRATEGY

A data governance system is one that regulates user access to personal information. Information on a particular, identifiable individual is referred to as individual data. An individual in such a system has rights, including the power to decide who has access to his or her information or to give approval to particular parties. Laws like the General Data Protection Regulation of the European Union may establish a data governance framework. In May 2018, the GDPR, a set of rules for data protection and privacy in the European Union, went into force. Patient data handling and processing in the UK is governed by the Data Protection Act (DPA) of 2018, which brought about the EU General Data Protection Regulation (GDPR). The rule guarantees that people have the right to access, update, erase, or restrict their personal data and mandates that data controllers get the express agreement of data subjects before collecting their personal information. The GDPR mandates that businesses implement security measures and disclose data breaches. Some of these safety measures include pseudonymization and the encryption of personal information (Guide to the UK General Data Protection Regulation (UK GDPR)). The GDPR is a crucial step in guaranteeing the moral use of big data and safeguarding of personal data in the healthcare sector. Beyond the information transmission and storage procedures employed by healthcare organizations, a safety concept applies. Every aspect of the management of personal data is covered. This means that the safeguarding measure in place should be effective in ensuring that the data can only be accessed, modified, shared, or removed by those who have been granted permission to do so; that the information a healthcare

organization holds is reliable and complete in relation to the reasons you are using it; and that the data is readily available and useful, meaning that if personal data is accidentally lost, changed or destroyed, the organization should be able to recover it. According to the GDPR, health organizations must adopt a degree of safeguarding that commensurate with the risks involved in their data processing operations. Organizations must assess and appreciate the potential dangers to the personal data they handle, as well as implement the required safeguarding measures, in order to manage these risks. Instead of mandating any specific safeguarding measures, the GDPR emphasizes the need for organizations to take a risk-based method to safeguarding or security. Organizations must make sure that patients are informed about how their information is utilized. A patient shouldn't be caught off guard by how their personal information is utilized (NHS choices). Every time personal information is transferred from one organization to another for purposes other than those for which the data subject is accountable, a data sharing contract should be in place.

The GDPR also imposes severe sanctions for non-compliance, including significant fines. When using big data in the healthcare sector, this acts as a potent deterrent for organizations to prioritize data protection and motivates them to follow ethical procedures. In addition to assisting in the protection of patient privacy, GDPR compliance promotes moral conduct among healthcare professionals, scientists, and other stakeholders.

6 CONCLUSION

Big data in the context of healthcare is a topic broadly discussed presently. The use of big data in healthcare is important because data from different sources can be used for prediction modelling, diagnosis, likelihood of a patient's developing a certain disease, medical research, financial management among others. The issue of patient privacy and data ownership has surfaced in the age of big data in healthcare but in my view, I believe if there is a strict compliance and adherence to the governance approach by GPDR for Data protection, there will be minimal ethical issues associated to the use of big data in healthcare industry. The GDPR encourages the appropriate and ethical use of big data in the healthcare sector for the benefit of patients and society at large thanks to its emphasis on strong data protection safeguards, individual rights, and responsibility.

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